Information System Department

King Saud University

College of Computer and Information Sciences

Information System Department

IS 346: Data Quality - Course Project

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Messy Clinic Appointment Report

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1. Introduction

This report outlines the implementation of a simple Data Quality Monitoring System to analyze a dataset and understand how data quality is monitored in real environments. The dataset chosen for this project is the "Messy Clinic Appointments" dataset from Kaggle, which simulates medical scheduling records.

Dataset Attributes Description:

• **patient_id**: Unique patient identifier.

• **patient_name**: Full name of the patient.

• **age**: Age of the patient.

• **gender**: Gender (Male/Female).

• **appointment_date**: Intended appointment date (dd/mm/yyyy).

• **booking_date**: Date the appointment was booked.

• **doctor**: Assigned doctor name.

• **department**: Medical department.

• **billing_amount**: Consultation fee (should be numeric).

• **follow_up_required**: Whether follow-up is needed (Yes/No).

2. Data Profiling

In this phase, we analyze the dataset and explore various statistical metrics. The dataset contains 1000 instances and 10 attributes.

• Minimum, maximum, and average values: The age attribute falls within expected biological limits, with a minimum of 18.0, a maximum of 90.0, and an average of 53.75.

• Missing values: The dataset contains missing values in two columns: billing_amount is missing 50 values (5.0%), and gender is missing 50 values (5.0%).

• Duplicates: There are 0 exact duplicate rows in the entire dataset. However, the patient_id column contains 899 duplicates.

• Outliers: Based on the boxplot generated for the age column using Matplotlib, no significant statistical outliers were detected.

3. Data Quality Issues Found

Based on the profiling analysis, we identified all data quality problems and classified them into four main dimensions:

• Completeness issues: The dataset is incomplete. There are missing values in critical columns, specifically 50 missing records for gender and 50 missing records for billing_amount.

• Uniqueness issues: The patient_id column, which is expected to be a unique identifier, contains 899 duplicate entries, indicating a severe primary key violation.

• Validity issues: There are significant validity problems:

• 707 records in the gender column do not strictly match 'Male' or 'Female' (containing noisy values like 0, 1, M, F).

• 950 records in the billing_amount column contain invalid text or currency symbols (e.g., €) making them non-numeric.

• Consistency issues: The dataset struggles with logical consistency:

• 908 records have an invalid appointment_date format that does not match the %d/%m/%Y standard.

• 16 records show future appointment dates, which may conflict with historical records.

• 672 records have inconsistent follow_up_required values that are not standard 'Yes' or 'No'.

4. Monitoring System Design

To track and maintain high data quality over time, a simple system to monitor data quality was designed.

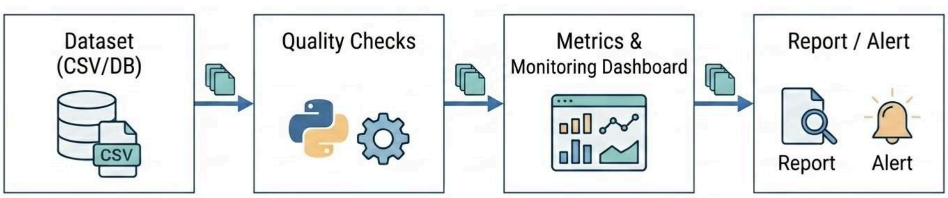
• What should we monitor? All incoming clinic appointment records generated by the system.

• What rules will we use? patient_id must be non-null and unique, age must be between 0-120, gender must strictly be 'Male' or 'Female', billing_amount must be purely numeric, and appointment_date must follow a valid date format.

• What metrics will we measure? We will measure the null count, unique duplicate rate, out-of-range rate, validity rate, and format error rate per attribute.

• When should alerts happen? Alerts should trigger in real-time upon data ingestion via batch/API, periodically through a weekly quality score dashboard, and automatically escalate via email to a data steward if thresholds are breached.

Diagram:



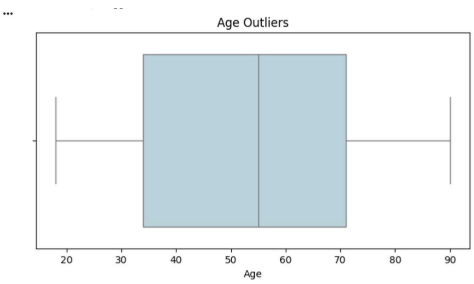
5.Results

```
DATASET DESCRIPTION
==================================================

Attribute            | Description
patient_id           | Unique patient identifier
patient_name         | Full name of the patient
age                  | Age of the patient
gender               | Gender (Male/Female)
appointment_date     | Intended appointment date (dd/mm/yyyy)
booking_date         | Date the appointment was booked
doctor               | Assigned doctor name
department           | Medical department
billing_amount       | Consultation fee (should be numeric)
follow_up_required   | Whether follow-up is needed (Yes/No)
```

This output acts as a data dictionary, detailing the column names (e.g., patient_id, age, department, billing_amount), the expected information for each, and their specific constraints (e.g., the consultation fee must be numeric).

```
... Name: count, dtype: int64

    --- doctor top values ---
    doctor
    Nancy Hernandez      2
    Nancy Wilson         2
    Wesley Johnson       2
    Kristen Lopez        2
    James Aguilar        2
    Christopher Smith    2
    Benjamin Brown       2
    Jennifer Adams       2
    Jennifer Carroll     2
    Michael Wright       2
    Name: count, dtype: int64
```

A statistical report showing the most frequently assigned doctors in the system. It displays the counts for specific names (like Nancy Hernandez and Wesley Johnson) to observe the distribution of appointments.



A box plot illustrating the distribution of patient ages. It shows that ages range approximately between 18 and 90 years, with no extreme outliers outside the logical range (0-120), indicating that the age data is valid and clean.

```
QUALITY CHECKS IMPLEMENTATION
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Note: No email column in dataset

--- Data Quality Check Summary ---
```

	Check	Failed Records
0	Billing Format (Currency)	950
1	Appt Date Invalid	924
2	Duplicate IDs	899
3	Gender Format	707
4	Follow-up Format	672
5	Missing Names	0
6	Age Range (0-120)	0
7	Billing Negative	0
8	Department Format	0
9	Email Format	0

A summary table listing the number of records that failed each specific quality check. It shows that billing formats and appointment dates have the highest failure rates, while checks for "Age Range," "Department Format," and "Missing Names" passed entirely with zero failures.

```
--- department top values ---
department
Neurology      273
Orthopedics    262
Cardiology     234
General        231
Name: count, dtype: int64

--- follow_up_required top values ---
follow_up_required
0      185
Y      182
Yes    181
N      154
1      151
No     147
Name: count, dtype: int64
```

Statistics showing the distribution of records across medical departments (Neurology leads with 273 cases). More importantly, it reveals a lack of standardization in the follow_up_required column (using mixed values like 0, 1, Y, N, Yes, No), indicating that data cleaning is required.

```
CLASSIFICATION OF DATA QUALITY ISSUES
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```
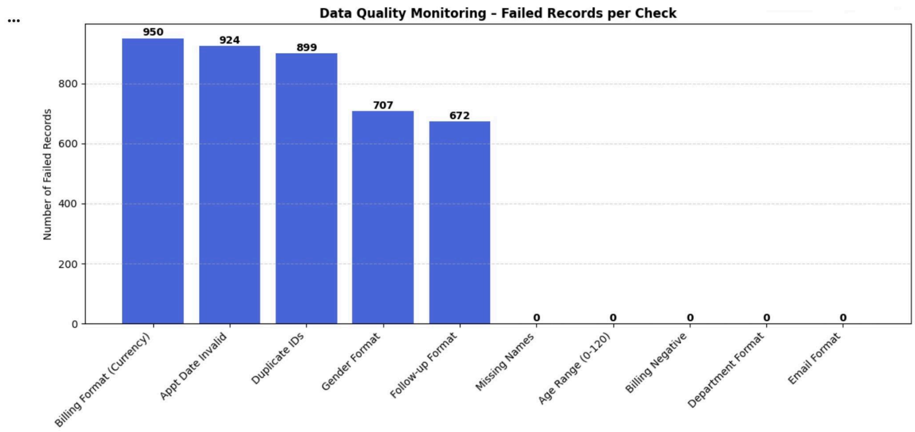

	Category	Issue Description	Extent
0	Completeness	Missing gender values	50 records
1	Completeness	Missing billing_amount values	50 records
2	Uniqueness	Duplicate patient IDs (should be unique)	899 duplicates
3	Validity	Gender not 'Male'/'Female'	707 records
4	Validity	Billing amount contains text/symbols	950 records
5	Validity	Age outside plausible range (0-120)	0 records
6	Consistency	Appointment date invalid format	908 records
7	Consistency	Future appointment dates	16 records
8	Consistency	Department values not in expected list	0 records
9	Consistency	Follow-up not Yes/No	672 records

A detailed table categorizing the discovered issues into standard data quality dimensions (Completeness, Uniqueness, Validity, Consistency). For example, it highlights 899 duplicate patient IDs and 950 errors where the billing amount contains text or symbols.



A visual representation of the number of failed records per quality check. This chart makes it easy to quickly identify that "Billing Format (Currency)" and "Appt Date Invalid" are currently the biggest data quality issues in the dataset.

6. Recommendations

To resolve the identified data quality problems, we recommend the following actions:

1.Data Cleansing Scripts: Apply a preprocessing function in Python using Pandas to strip out currency symbols from the billing_amount column and cast it to a standard numeric type.

2.Input Standardization: Implement strict drop-down menus in the booking application's user interface to restrict the gender field to 'Male' or 'Female', and the follow_up_required field to 'Yes' or 'No'.

3.Investigate ID Generation: The high number of duplicate patient_ids suggests a system error. The database team must investigate whether these represent multiple appointments for the same patient, in which case a new composite key (e.g., appointment_id) must be introduced to ensure table uniqueness.